

CSCI 315 – Assignment 5

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	Epochs	Learning Rate	Accuracy	Training Time (s)	Test Loss
CNN CPU	40	0.01	98.706%	1719.732	0.00038592
CNN GPU	40	0.01	98.627%	143.325	0.00040049
CNN GPU	500	0.01	98.676%	1810.070	0.00069572
MLP	500	0.01	97.681%	640.875	0.0010223
SLP	500	0.01	92.108%	581.623	0.0041191

Discussion:

Although the GPU reduced CNN training time by a rough factor of 10 (from 29 minutes -> 2.5 minutes), I actually found that the accuracy was very similar. However, if there were larger discrepancies in accuracy, one reason might be that GPUs tend to use lower floating point accuracy for speed. This minor difference might accumulate over many epochs. Another potential reason is the possible non-determinism of parallel computing.

Seeing that a larger learning rate performed quite well for the MNIST dataset in the previous assignment, I used a learning rate of 0.01 instead of 0.0001.

For the GPU, I used the RTX 5090 32vCPU through RunPod (<https://www.runpod.io/>) (I apologize, the JetStream2 allocation was closed due to my late submission, so I found an alternative)

Compared to MLP and SLP for the same number of epochs and learning rate, the CNN performed better. The CNN averaged a percentage higher in accuracy, a considerable gain.

Below are the confusion matrices for each respective models / hyperparameters:

CNN CPU; 40 epochs

Actual Digit	Classification Output									
	0	1	2	3	4	5	6	7	8	9
0	977	0	0	0	0	0	1	1	1	0
1	0	1128	1	0	1	1	1	2	1	0
2	1	2	1024	0	0	0	0	3	2	0
3	1	0	1	1002	0	3	0	1	2	0
4	0	0	1	0	977	0	1	0	0	3
5	2	0	1	4	0	882	1	0	1	1
6	2	2	0	1	1	6	946	0	0	0
7	0	1	4	1	0	0	0	1020	1	1
8	1	0	1	3	0	1	0	1	964	3
9	0	0	0	0	5	3	0	3	0	998

CNN GPU; 40 epochs

Actual Digit	Classification Output									
	0	1	2	3	4	5	6	7	8	9
0	976	0	0	0	0	0	1	1	2	0
1	0	1132	1	0	0	1	0	1	0	0
2	1	4	1019	0	2	0	0	2	4	0
3	1	0	1	999	0	6	0	1	2	0
4	0	0	1	0	976	0	1	0	1	3
5	2	0	1	3	0	883	1	0	1	1
6	0	2	0	1	1	8	946	0	0	0
7	0	2	3	1	0	0	0	1019	1	2
8	3	0	1	1	0	1	0	1	965	2
9	0	2	0	0	4	3	0	4	1	995

CNN GPU; 500 epochs

Actual Digit	Classification Output									
	0	1	2	3	4	5	6	7	8	9
0	978	0	0	0	0	0	0	0	1	1
1	0	1133	1	0	0	0	0	0	1	0
2	1	1	1021	0	1	0	1	3	4	0
3	1	0	1	1003	0	4	0	0	1	0
4	0	0	0	0	974	0	1	0	0	7
5	2	0	0	5	0	882	1	1	0	1
6	2	2	0	1	2	6	943	0	2	0
7	0	1	2	1	0	0	0	1022	1	1
8	2	0	2	1	0	1	0	1	965	2
9	1	0	0	2	7	3	0	2	0	994

MLP; 500 epochs

Actual Digit	Classification Output									
	0	1	2	3	4	5	6	7	8	9
0	969	0	1	0	1	2	2	2	2	1
1	0	1125	3	1	0	1	2	1	2	0
2	3	2	1009	4	3	0	3	4	4	0
3	0	0	4	992	0	3	0	3	1	7
4	1	0	2	1	963	0	3	2	1	9
5	2	0	0	6	1	874	4	1	2	2
6	4	2	1	1	2	3	944	0	1	0
7	1	3	7	2	0	0	0	1005	3	7
8	4	1	3	4	2	1	2	3	952	2
9	1	2	0	4	9	2	1	5	3	982

SLP; 500 epochs

Actual Digit	Classification Output									
	0	1	2	3	4	5	6	7	8	9
0	958	0	2	3	0	7	5	4	1	0
1	0	1111	3	2	0	2	4	2	11	0
2	6	9	933	15	8	3	13	8	30	7
3	4	1	17	919	0	24	3	11	22	9
4	1	2	6	3	909	0	8	3	9	41
5	10	2	3	36	9	775	16	6	30	5
6	10	3	8	1	8	14	910	2	2	0
7	1	7	23	7	8	1	0	949	2	30
8	8	11	7	23	8	25	10	8	862	12
9	10	8	1	9	22	5	0	17	8	929